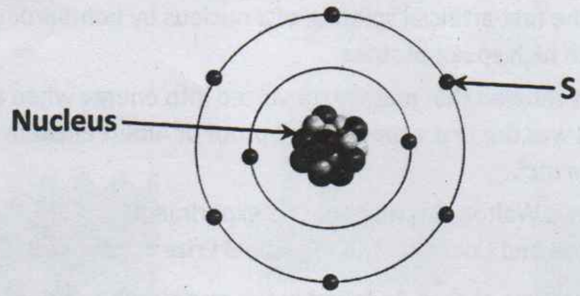
Question 1 — 15 marks

Atoms contain subatomic particles. The diagram below represents the structure of the atom proposed by Niels Bohr in 1913.

95%



(a) (i) In Bohr's model, subatomic particle **S** orbits the nucleus and has a negative charge. What is the name of this particle? Put a tick (✓) in the correct box.

Electron ☑ Neutron ☐ Proton ☐

✓

(ii) Which subatomic particle is located in the nucleus of an atom and has no charge? Put a tick (✓) in the correct box.

Electron ☐ Neutron ☑ Proton ☐

✓ 4

(iii) Bohr's model of the atom is often referred to as a 'planetary model'. Suggest a reason for this.

The electrons orbit the nucleus like the planets of the Solar system orbit the sun

✓ 2

iv)

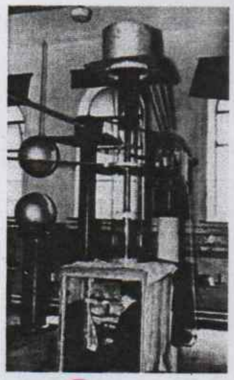
Match each of the following sub-atomic particles to their descriptions in the table below.

~~Electron~~ ~~Neutron~~ ~~Proton~~

Description	Particle
Positively charged	Proton ✓
Negatively charged	Electron ✓
No charge	Neutral ✓

9

Question 2 15 marks



In 1932, the Irish physicist Ernest Walton and the English physicist John Cockroft produced the first artificial splitting of a nucleus by bombarding atoms of lithium with high speed protons.

Walton and Cockroft showed that mass is converted into energy when a nucleus is split. This was the first experimental proof of Albert Einstein's famous equation, $E = mc^2$.

The photograph shows Walton carrying out this experiment.
For their work, Walton and Cockroft won the Nobel Prize in physics in 1951.

(a) Protons are one type of particle found in the nucleus of an atom.

(i) Name the other type of particle found in the nucleus of an atom.

Neutrons

3

(ii) Compare the charge of the proton with the charge of the other particle you have named.

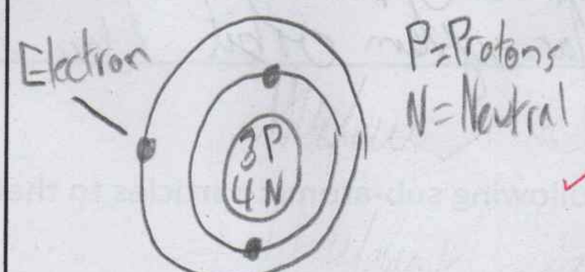
The charge of the proton is positive compared to the neutron's neutral charge

3

(b) In the space below, draw a labelled diagram of an atom of lithium showing the positions of the nucleus and the electrons.

Li 3 7

Lithium atom



6

The work of Walton and Cockroft saw the dawn of the nuclear age.

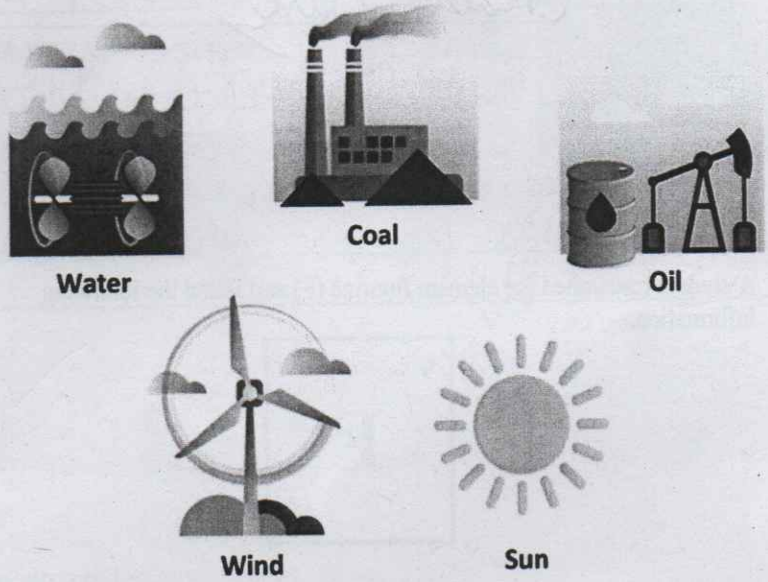
(c) State one advantage of using nuclear power to generate electricity.

3

It is very efficient, it generates a lot of electricity

Question 3 — 14 marks

The images below show sources used by humans to generate electricity.



(a) Classify each of the sources as renewable or non-renewable by putting a tick (✓) in the correct column.

Source	Renewable	Non-renewable
Water	✓	
Coal		✓
Oil		✓
Wind	✓	
Sun	✓	

10

(b) "Energy producers face ethical issues in the way energy is extracted and/or produced." Do you agree with this statement? Justify your answer.

Yes because to get energy you have to take it from somewhere and that often harms the local environment through noise pollution or the release of chemicals

4

Question 4

4 marks

(*f*) Give **one *advantage*** and **one *disadvantage*** of using nuclear energy to generate electricity.

Advantage Efficiency

Disadvantage Creates hazardous waste

4

Question 5

14 marks

use only

(24) (1) (2)

(*c*) A student researched the element fluorine (F) and found the following information.

9	
F	
	19

(*i*) Name one particular source which the student could have used to find this information. Periodic table

2

(*ii*) Using this information, show in the table below the *number* of each type of sub-atomic particle present in an atom of fluorine. Complete the table to show the *location* (within the atom) of the protons and of the neutrons.

Particle	Number	Location
Proton	9	Nucleus 4
Neutron	10	Nucleus 4
Electron	9 2	Electron cloud

2,7

2+8+7=17

(*iii*) Name an element which has the same number of electrons in its outer shell as fluorine. Chlorine

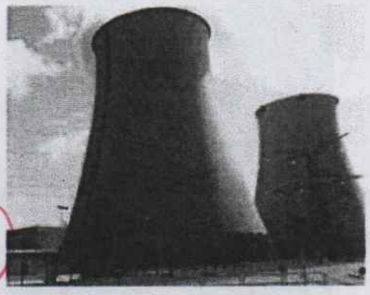
2

Question 6 — 6 marks

(*h*) France produces most of its electricity using nuclear sources, such as uranium.

(*i*) Are nuclear energy sources considered to be *renewable* or *non-renewable* sources of energy?

Non-renewable



Explain your answer. There is a finite amount and eventually it will run out if we keep extracting it 3

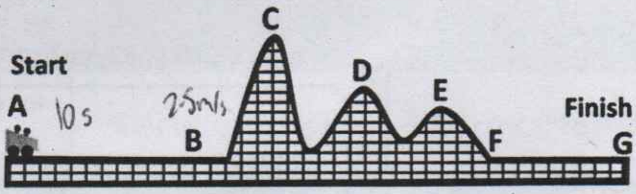
(*ii*) State one advantage of producing electricity using nuclear energy sources.

Nuclear energy is very efficient 3

Question 7 — 3 marks

The diagram shows a rollercoaster car at the start of its track. The letters **A** to **G** indicate the position of the car at certain stages during its journey along the track.

The car is stationary (not moving) at point **A**. It takes 10 seconds for the car to reach point **B**. As the car passes point **B** it is travelling at a speed of 2.5 m/s.



(c) Which letter represents the point on the track when the car has the most potential energy? Justify your answer.

C because it is the highest and can drop the furthest so has the most potential energy 3

Question 8 — 3 marks

(*c*) Name each of the metallic elements with the chemical symbols below.

Au Gold

Cu Copper

Fe Iron

3

Question 9   18 marks

Sankey diagrams are named after H. Riall Sankey, a Tipperary-born engineer, following his 1898 description of the energy efficiency of a steam engine.

Sankey diagrams show the flow of energy to and from a device.

In a Sankey diagram, the width of each arrow represents the energy named.

The Sankey diagrams for a filament lamp and a compact fluorescent lamp (CFL) are shown below.

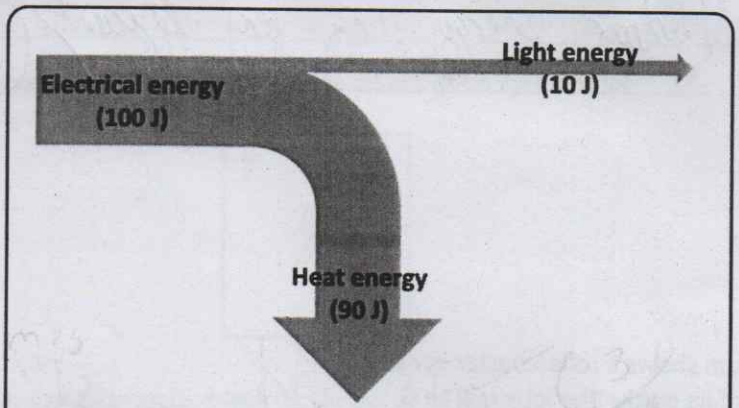


Figure 1 Sankey diagram for a filament lamp

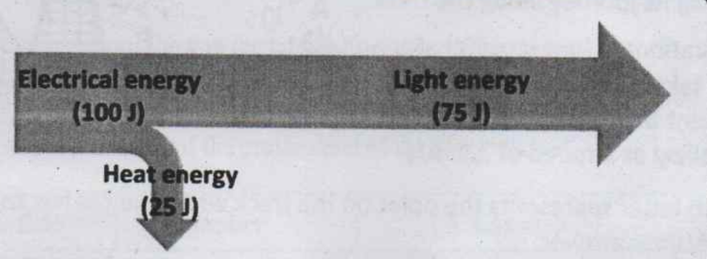


Figure 2 Sankey diagram for a compact fluorescent lamp (CFL)

(a) Examine figures **1** and **2**. Which lamp is more efficient? Justify your answer.

Figure 2 is more efficient because more of the energy put in is going where it is wanted to

2

Calculate the energy efficiency of both figure 1 and figure 2

$\frac{10}{100} \times 100 = 10\%$ for Figure 1

$\frac{75}{100} \times 100 = 75\%$ for Figure 2

4

(b) What is the law of conservation of energy?

Energy cannot be created or destroyed, just changed from one form to another ✓ 2

(c) A student is asked to investigate and compare the heat energy produced by filament lamps and CFLs.

Apart from the lamps themselves, name a piece of equipment that could be used during this investigation. Explain how this piece of equipment could be used during the investigation.

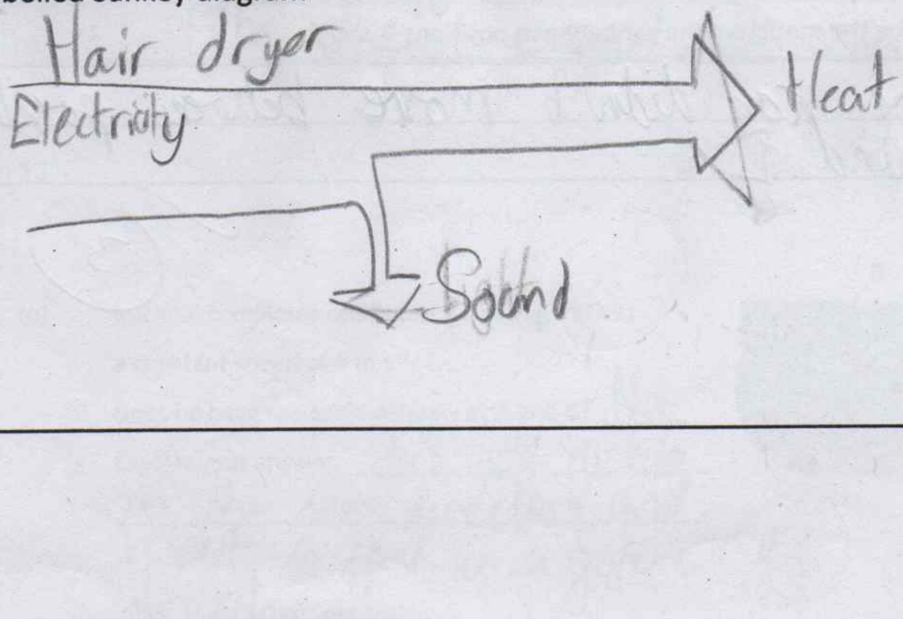
A thermometer could be used to measure how hot the lamps are and compare the different ~~heat~~ temperatures ✓ 3

(d) The energy conversions that happen in a CFL are described in the table below.
Complete the table for another device which transforms energy from one form to another and which you designed as part of your studies in science. Your design!

Name of the device	Function of the device	Main useful energy conversion	Main loss of energy
Compact fluorescent lamp (CFL)	To provide artificial light	Electrical to light	Electrical to heat
Hair dryer	To heat air and blow it	Electrical to heat	Electrical to sound

(e) Sketch a Sankey diagram for the device you described in part **(d)**. Your design!
Label each part of the diagram.

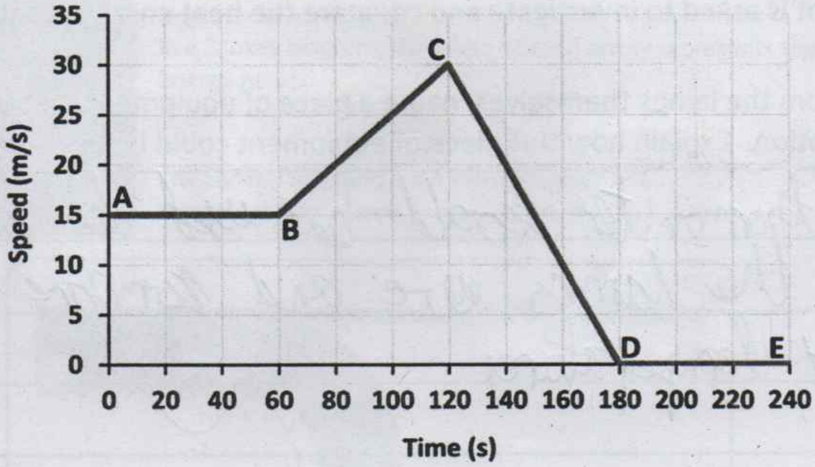
Labelled Sankey diagram



Question 10

12 marl

The graph below shows how the speed of a car changed with time during a journey.



(a) What is the maximum speed of the car during this journey?

30 m/s

3

(b) Calculate the acceleration of the car between positions **B** and **C**.

Calculation

30 − 15 = 15

120 − 60 = 60

15 ÷ 60 = 0·25

Answer = 0·25 m/s/s

3

(c) Describe the motion of the car between positions **C** and **D**.

The car slowed down to a halt between positions C and D

3

(d) Describe the motion of the car between positions **D** and **E**.

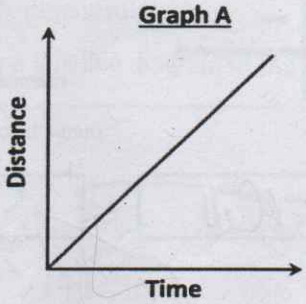
The car didn't move between positions D and E

3

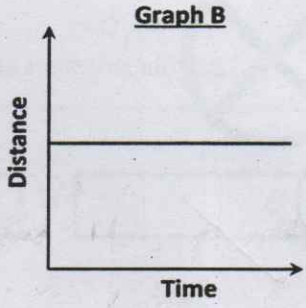
Question 11 — 12 marks

The four graphs below (**A, B, C** and **D**) describe the motion of four different objects.

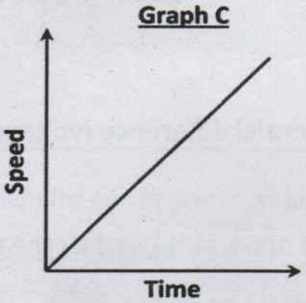
Graph A



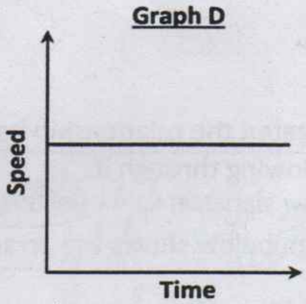
Graph B



Graph C



Graph D



Use the statements below to describe the motion of the object in each of the four graphs.
(Note that one of the statements should be used twice.)

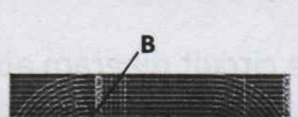
1. The object did not move.
2. The object moved with constant speed.
3. The object accelerated.

Graph	Statement
A	The Object moved with a constant speed
B	The Object did not move
C	The object accelerated
D	The Object moved with a constant speed

12

Question 12 — 6 marks

(*b*) Micheál completes one lap of a running track at a constant speed of 6 m s^{-1}.



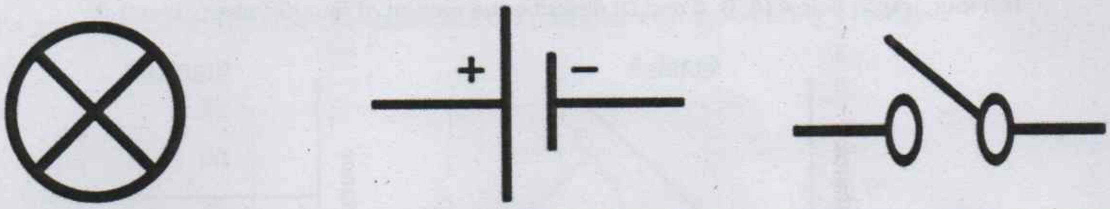
(*i*) Does he have the same velocity at **A** and **B**? No

(*ii*) Explain your answer. He's not running in the same direction and velocity is speed in a given direction

6 Excellent!

Question 13 — 6 marks

(a) In relation to circuits, identify the following symbols.



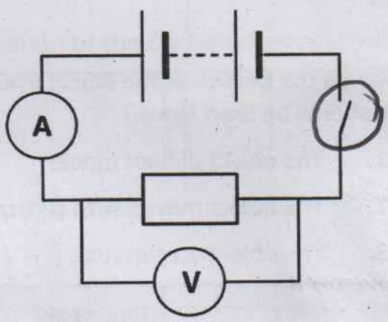
Bulb	Battery / Cell	Switch

6

Question 14 — 4 marks

A student investigated the relationship between the potential difference (voltage) across a resistor and the current flowing through it.

The circuit diagram below shows the arrangement of the apparatus used by the student.



1

Examine the circuit diagram and answer the questions below.

(a) The instrument labelled **V** measures voltage. Name instrument **V**.

Voltmeter

3

(b) The instrument labelled **A** measures current. Name instrument **A**.

Ammeter

(c) In the circuit diagram above, draw a circle around the symbol for the switch.

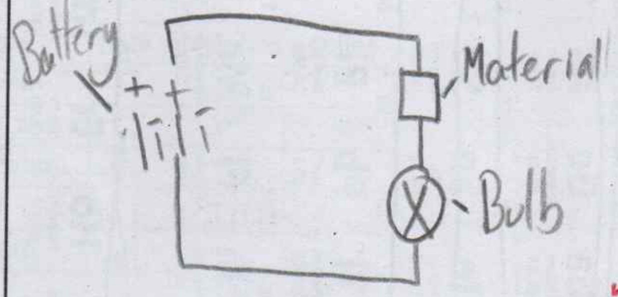
Question 15

9 marks

(*b*) A student was asked to test electrical conduction in *paper*, *copper*, *silver* and *plastic* and conclude which of these materials were electrical conductors and which were insulators. (9)

(*i*) Draw a labelled diagram of the set up (circuit) the student could use.

Labelled diagram



6

(*ii*) Based on the circuit you drew above, how could you tell which materials were electrical conductors? If the bulb lights, electricity is flowing through the material, making it a conductor

3

Question 16

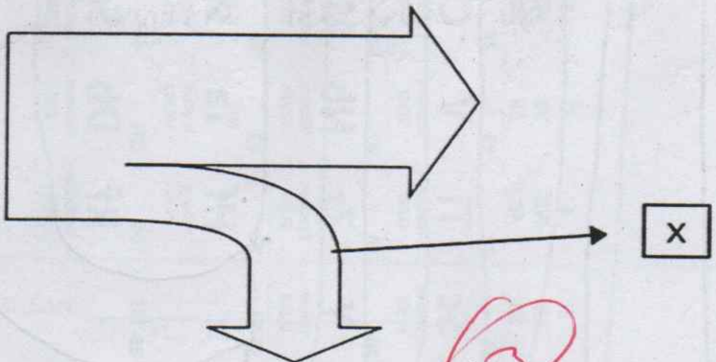
9 marks

(b) What are the units of measurement for energy?

3

Joules

(c) What name is given to the diagram below to represent energy transfer?



3

Sankey diagram

(d) What term is used to describe the energy at **X**?

3

Dissipated energy